# Fed Signal Chatbot: Grounding LLMs in Federal Reserve Communications via Hybrid Retrieval

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## 1. Introduction

Understanding the Federal Reserve's monetary policy stance is essential for anyone interested in financial markets and stocks, yet interpreting Federal Reserve Open Market Committee (FOMC) communications is not so simple. It requires reading dense financial jargon which is a difficult task for the average retail investor. When I first started learning about stock market trading, I didn't understand why professional traders reacted so violently to specific phrases. I didn't understand the memes about how Federal Reserve Chair Jerome Powell's greeting at the FOMC meeting can dictate the market. I had to learn the hard way when I saw the direct correlation between the Fed's rhetoric and my account balance crashing.

This project asks whether an LLM can give a grounded answer when a user asks what the Fed is currently doing. General knowledge models have two failure modes on this task: they hallucinate on meetings outside their training cutoff, and on memorized historical meetings they answer from recall rather than inference, making it impossible to distinguish reasoning from memorization. We address both with a retrieval-augmented (RAG) system and split the golden test set by the LLM's training cutoff so the baseline cannot fall back on priors. We also report retrieval quality with a binary hit@5 metric alongside accuracy, because as section 5.3 shows, accuracy alone can rise on a class-imbalanced split even when retrieval is failing.

## 2. Context and Related Work

Important context that should be known for this project is that analysts and economists usually classify each Fed meeting with a stance of: Hawkish (indicating they are trying to fight high inflation), Dovish (indicating they are focusing on economic growth), and Neutral (a middle ground between the two previous stances).

Retrieval-augmented generation grounds LLM outputs in retrieved documents (Lewis et al., 2020). Our system combines lexical retrieval (Robertson and Walker, 1994), dense sentence-embedding retrieval (Reimers and Gurevych, 2019), and Reciprocal Rank Fusion (Cormack et al., 2009).

FOMC text analysis has a long history in economics. Lucca and Trebbi (2009) introduced semantic-orientation scoring of FOMC statements and showed that text-derived hawkishness measures correlate with subsequent policy moves. More recently, fine-tuned BERT-style classifiers like FinBERT (Araci, 2019) have been applied to sentence-level stance classification of Fed text.

## 3. Approach

### 3.1 Corpus and ingestion

We scrape all FOMC statements published by the Federal Reserve from 2006-1-31 onward using Beautiful Soup, yielding 175 statements at the time of creating this project. The Federal Reserve makes these statements freely available as part of its public communication mandate, and they are works of the U.S. federal government and therefore in the public domain. Statement URLs are discovered by regex from fomccalendars.htm

and fomchistorical{YYYY}.htm, with two URL path schemes handled (monetary{YYYYMMDD}a.htm for 2011 onward and monetary/{YYYYMMDD}{a-z}.htm for 2006-2010).

### 3.2 Chunking and embedding

Each statement is split into 256-token chunks with 32-token overlap. The chunk size is chosen to match the maximum sequence length of the all-MiniLM-L6-v2 sentence-transformer (Wang et al., 2020) used as the embedder, so that no chunk is silently truncated at embed time. The 32-token overlap exists because Fed statements often split a single idea across paragraph breaks (for example, the rate decision may appear in one paragraph and the forward guidance in the next); without overlap, retrieval can surface one half without the other. Chunks are embedded with all-MiniLM-L6-v2, L2-normalized, and stored in ChromaDB (Chroma, 2023) with cosine similarity and metadata fields ({meeting_date, source_url}). The same chunks are tokenized at the word level with regex, and indexed with rank-bm25 BM25Okapi. We use whole words rather than the MiniLM subword tokenizer because BM25's IDF weighting is undefined on subword fragments. The total index covers 572 chunks.

### 3.3 Retrieval

We use three retrieval modes:

1. Semantic-only: top-k by cosine similarity on the dense embedding
2. BM25-only: top-k by BM25 score over the same chunks
3. Hybrid: each retrieval form from above produces a candidate pool of 20. Those candidates are scored by Reciprocal Rank Fusion, $RRF(c) = 1/(k + rank_{semantic}(c)) + 1/(k + rank_{BM25}(c)),\ k = 60,$ and the global top=5 is returned. The constant k=60 is the value most commonly cited in the literature (Cormack et al., 2009) and we did not change it. All three modes return exactly five chunks to the generator, so the only thing varying across the ablation is which five chunks the generator sees.

### 3.4 Generation

The generator is GPT-4o-mini accessed through the OpenAI API in JSON mode at temperature 0. The prompt has four parts: (1) a system message scoping the model to FOMC stance classification with a “no investment advice” guardrail, (2) three few-shot examples, one each of hawkish (2022-06-15), dovish (2020-03-15), and neutral (2016-09-21), with retrieved snippets quoted verbatim from the corresponding raw statements, (3) the retrieved context with each chunk's meeting date as a header, and (4) a chain-of-thought instruction (Wei et al., 2022) asking the model to reason through three steps (what language signals stance, what does the retrieved context say, what is the classification) before committing. The output is parsed against the closed set {hawkish, neutral, dovish}, and any output that does not parse or that explicitly refuses is recorded as 'unknown'.

### 3.5 Design Tradeoffs

**Embedder:** We use all-MiniLM-L6-v2 (384-dim, 256-token context) over stronger alternatives like bge-large-en-v1.5 (1024-dim, 512-token) or E5-large for one main reason. MiniLM runs comfortably on CPU at zero per-call cost, while bge-large requires either a GPU or a paid embedding API. Second, FOMC statements are short (typically only 600-900 tokens) and produce only 2-5 chunks each. A stronger embedder on this corpus is unlikely to drastically improve accuracy, and thus does not justify its compute cost.

**Generator:** We use GPT-4o-mini over a stronger model (GPT-5, Claude Opus) and over local alternatives (Llama 3 8B-instruct, Mistral 7B). The stronger commercial models cost roughly 15-30x more per call and were not the bottleneck in our pipeline. Preliminary tests showed retrieval, not generation, was the main driver of accuracy on this task. Local models would eliminate the API cost and data-privacy concerns, but since the corpus is public-domain US government text there is no privacy risk from sending it to OpenAI.

**Figure 1:** The four-stage pipeline of the model

**Reproducibility:** The numbers reported in this paper come from "gpt-4o-mini-2024-07-18" at temperature 0, JSON mode, with seed=42. OpenAI's system_fingerprint is logged on every call and surfaced in the answer() function's return value. A fingerprint change indicates that OpenAI altered backend state, in which case the seed no longer reproduces prior outputs and the evaluation should be re-run.

## 4. Evaluation Design

The main methodological choice in this project was to split the golden test set by the LLM's training cutoff, which we set to 2023-10-01 to match GPT-4o-mini's approximate cutoff: pre-cutoff era and post-cutoff era. The motivation is that previous FOMC statements are present in the model's pretraining data. That means, if we were to report a single overall accuracy number, the model's pretraining recall on those meeting inflates the no-retrieval baseline and makes the contribution of retrieval look small.

### 4.1 Golden test set

The set contains 38 meetings hand-labeled from the raw statement text. Pre-cutoff has 26 meetings spanning 2006-2023 with class balance 10 hawkish / 8 dovish / 8 neutral. Post-cutoff has 12 meetings spanning 2024-2026 with class balance 0 hawkish / 4 dovish / 8 neutral. The reason we have 0 hawkish examples post-cutoff is because there were no hawkish leaning statements since 2023-07-26. This is of course a limitation, but is not a massive barrier in our analysis. Also, the three meetings in our few-shot examples are excluded from the test set so that we avoid any train/test leakage.

### 4.2 Query template and metrics

For each meeting, we run the query "*Classify the Fed's stance at the FOMC meeting on {meeting_date}.*" in each of the four modes (baseline, semantic-only, BM25-only, hybrid). The template is simple and serves the purpose. For each retrieval method, we report the following metrics: stance accuracy (fraction of correctly classified meetings), Hit@5 (the primary retrieval metric which indicates whether the top-5 retrieved chunks include any chunk from the target meeting (binary)), Precision@5 (the secondary metric which represents the fraction of the top-5 chunks whose meeting_date equals the target), and invalid-stance rate (which represents the fraction of unknown classifications).

# 5. Results

**Table 1.** Tier 1 stance ablation: 38 meetings × 4 modes = 152 GPT-4o-mini calls. *Hit@5* is the primary retrieval metric; baseline has no retrieval and so no hit@5. **Bold** marks the per-era best on each metric.

| Mode | Pre-cutoff (n=26) | | Post-cutoff (n=12) | | Overall (n=38) | |
|---|---|---|---|---|---|---|
| | Acc | Hit@5 | Acc | Hit@5 | Acc | Hit@5 |
| Baseline (no retrieval) | 0.731 | n/a | 0.000 | n/a | 0.500 | n/a |
| Semantic-only | 0.077 | 0.000 | 0.583 | 0.000 | 0.237 | 0.000 |
| BM25-only | **0.731** | **0.962** | **0.917** | **0.917** | **0.789** | **0.947** |
| Hybrid (RRF) | 0.692 | 0.769 | 0.750 | 0.750 | 0.711 | 0.763 |

## 5.1 The era-split difference

The headline finding is that the no-retrieval baseline behaves completely differently on the two eras. On pre-cutoff meetings it reached 73.1% accuracy, and on post-cutoff meetings it scores 0.0% (12/12 refusals). After inspecting the model's reasoning on pre-cutoff meetings, we see that it's recalling the actual statement text from pretraining and not from inference. This indicates that the 73.1% number measures memorization, not reasoning. The post-cutoff bracket, where the model has no priors, is the clean measurement of retrieval's contribution. BM25 retrieval converts the baseline's 0.0% into 91.7% on the same 12 meetings (Figure 2).

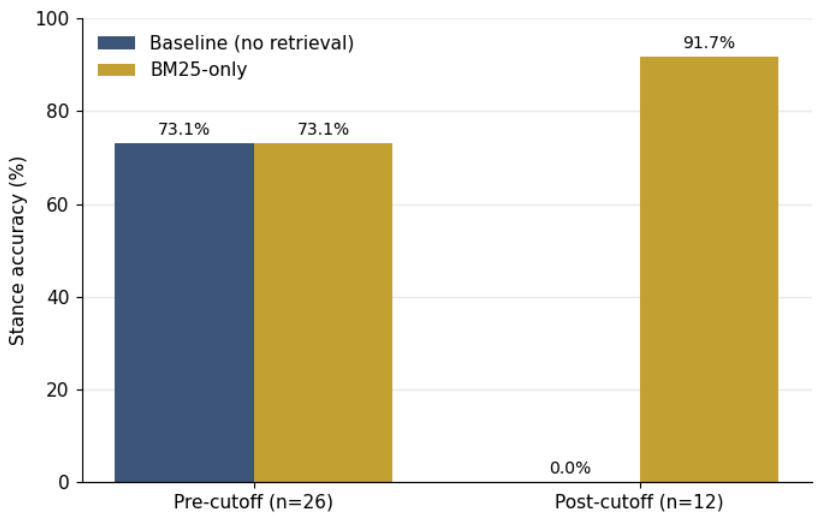


**Figure 2:** Stance accuracy results between baseline and BM25 retrieval in the different eras.

## 5.2 Hybrid retrieval underperforms BM25

The result that surprised me was that hybrid retrieval, which is usually considered the "best of both worlds" default in much of the RAG literature, lost to BM25 alone on this task. Post-cutoff stance accuracy is 75.0% for hybrid versus 91.7% for BM25, and the hit@5 gap goes the same direction (0.75 vs 0.92). This is because RRF assumes that both retrieval modes reinforce each other, so any chunk that ranks well in both is boosted strongly. Since our query is essentially just a date, the semantic retrieval has no useful representation of the date, so it ranks semantically related but wrong meetings highly. This therefore displaces the BM25-correct meeting from the fused top-5.

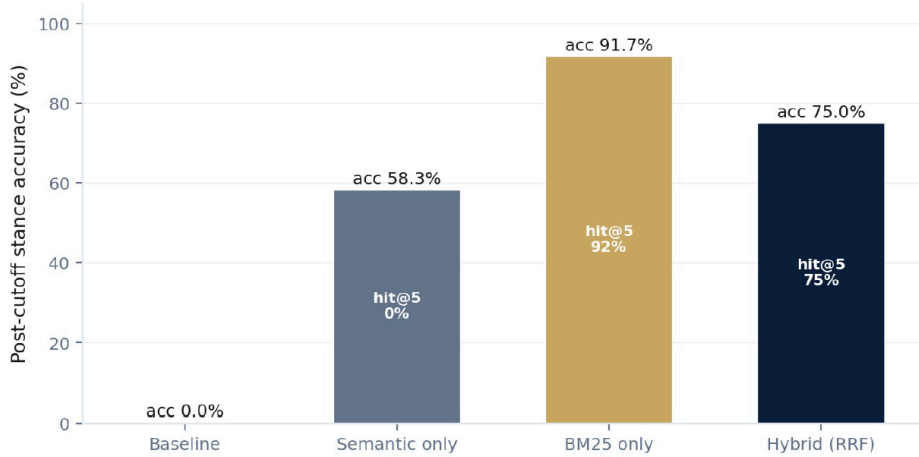


**Figure 3:** Post cutoff stance accuracy and hit@5 by mode.

### 5.3 Semantic retrieval performs poorly on metadata-style queries

Semantic-only retrieval returned hit@5 = 0.000 in both eras, meaning the dense embedder never surfaced a chunk from the target meeting across all 38 queries. Our query template has only one content-bearing token, the date, which is just a token sequence in a vector space populated by stance-related content. The takeaway is that for metadata-style queries, semantic retrieval is insufficient on its own and a keyword or structured retrieval path is required. What makes this a useful diagnostic is that semantic-only's accuracy still jumps from 7.7% pre-cutoff to 58.3% post-cutoff, which looks like a real improvement but isn't. The post-cutoff set has 0 hawkish samples (0H/4D/8N), so a model that tends to guess "neutral" or "dovish" lands correct more often than in the more balanced 10H/8D/8N pre-cutoff set (see Table 2). Since hit@5 doesn't depend on label distribution, it's the trustworthy retrieval signal here.

**Table 2.** Per-class confusion (post-cutoff, n=12). Rows are ground truth, columns are predictions. The post-cutoff bucket has 0 hawkish meetings by construction (the Fed has been on hold or cutting since July 2023). Baseline's "unknown" column is 12/12 refusals. BM25's only error is one dovish meeting predicted as neutral.

| Mode · Truth→ | Hawk | Neut | Dov | Unkn |
|---|---|---|---|---|
| **Baseline** · Neutral (n=8) | 0 | 0 | 0 | 8 |
| · Dovish (n=4) | 0 | 0 | 0 | 4 |
| **Semantic** · Neutral (n=8) | 0 | 6 | 1 | 1 |
| · Dovish (n=4) | 0 | 3 | 1 | 0 |
| **BM25** · Neutral (n=8) | 0 | 8 | 0 | 0 |
| · Dovish (n=4) | 0 | 1 | 3 | 0 |
| **Hybrid** · Neutral (n=8) | 0 | 8 | 0 | 0 |
| · Dovish (n=4) | 0 | 3 | 1 | 0 |

### 5.4 Error analysis

To understand where BM25 still fails, we manually inspected all 8 of its misclassifications across the 38 meetings. Only one was a retrieval miss (2008-03-18, where BM25 surfaced unrelated meetings and the model correctly abstained). The other 7 cases retrieved the right meeting but the model read it wrong, and 5 of those followed the same pattern: the model fixated on generic, cautious phrasing like "highly attentive to inflation risks" or "risks are roughly in balance" and under-weighted the actual rate decision. The clearest example is the only post-cutoff error, 2024-11-07, where the chunk contains an explicit 25 bp (basis points) cut but the model treated the dual-mandate hedging as the dominant signal and called it neutral. This shows that with BM25 surfacing the right document over 95% of the time, the remaining accuracy gap is mostly a reading problem, not a retrieval one.

## 6. Discussion

Several design choices in this project were not the first thing I tried. I documented what didn't work and reflected on those decisions here:

**Precision@5 was the original primary retrieval metric:** The first time I ran my evaluation, I only reported precision@5 alongside accuracy. BM25's mean precision@5 was 0.20 even though it was getting many examples right. I noticed that each FOMC statement chunks into only 2-5 chunks, so the precision@5 ceiling for a meeting that has three chunks is 0.60 even with perfect retrieval. This led me to finding different retrieval metrics for RAG models to get a more accurate representation of which mode is actually working and which isn't. We already discussed before why accuracy alone isn't enough (model could be falling back on priors). That's when I replaced precision@5 with hit@5 as the primary signal and kept precision@5 as a secondary ranking-quality metric.

**The original golden test set was 28 rows without a split:** An earlier version of my evaluation used 28 hand-labeled meetings without the era split, on which baseline reached 61%. I initially thought that the task was solvable from priors. But when I looked deeper into the results, I realized that the model's reasoning was clearly recalling memorized statement text instead of inferring. That's when I tried to split the evaluation into two groups, and added some more samples.

**I tried a Tier 2 evaluation with open-ended queries:** After finishing the Tier 1 ablation (this project), I built a small Tier 2 set of 20 open-ended informational queries (e.g., "How did the Fed's stance shift between July and September 2025?") and ran them through an LLM-as-a-judge pipeline. The numbers were much weaker than Tier 1 (best citation recall ~0.25) and the same RRF failure mode showed up again, this time on period-style queries that name two meetings. With more time, the extension would be to grow this set, build a stronger non-RRF retriever for multi-meeting queries, and replace the LLM judge with human review on a sample. I scoped Tier 2 out of the headline evaluation because its results were noisier than Tier 1's and would have made the paper's main claim about retrieval harder to read cleanly. For fun, I deployed the informational chatbot here: https://huggingface.co/spaces/hdawy/fed-signal-chatbot

## 7. Ethics & Limitations

### 7.1 User-facing risks and stance ambiguity

The system classifies stance and surfaces source meetings; it should not be recommending trades. The system prompt explicitly forbids investment advice and the live deployment carries a "not financial advice" caption, because a user might read a confident "hawkish" classification and infer a view on rates, equities, or bonds. Even with retrieval, the generator can still hallucinate or interpret hedged language too aggressively, and citations don't guarantee the cited meeting actually contains the claim. Stance itself is also ambiguous on borderline meetings where economists routinely disagree, which means my hand-labeled golden set reflects author judgment rather than consensus.

### 7.2 Post-cutoff class imbalance

Mentioned earlier, the post-cutoff bucket has zero hawkish meetings because the Fed's last hike (hawkish meeting) was in 2023 before the training cutoff. A model that blindly predicts just "neutral" or "dovish" will perform better in the post-cutoff set. This is mitigated by reporting hit@5, which does not depend on labels, as the primary retrieval metric.

### 7.3 Generation is the remaining bottleneck

As section 5.4 shows, most of BM25's errors are reading failures, not retrieval failures. That means our headline accuracy is bounded by GPT-4o-mini's stance-reading on this prompt rather than by retrieval, and a stronger generator or a more explicit prompt would likely close most of the remaining gap. The numbers here should be read as a lower bound on what's achievable on this corpus.

## 8. Future Work

Another type of extension could be testing out different chunking methods and retrieval methods. Some alternative chunking methods I could explore instead of fixed size token chunks could be paragraph based chunks, or heading based chunks. And for alternative retrieval methods, we would be interested in exploring how HyDE (Gao et al., 2023), an information retrieval technique that uses an LLM to generate a hypothetical answer to a query, can improve retrieval for each of the three modes (especially semantic).

The Tier 2 informational chatbot would be another possible development for future work. It would consist of three steps. First, the Tier 2 informational chatbot I described in section 6 could be developed into a primary evaluation for itself, with a larger test set, an LLM as a judge evaluator, and a richer multi-target retrieval to handle period-style queries. Second, retrieval over FRED (Federal Reserve Economic Data) macro features rather than text, like finding meetings where DGS10 (10 year treasury bond) was around 4% and unemployment was rising. Third, a relative-date query rewriter that would handle phrases like “the most recent meeting” or “two meetings ago”.

## 9. Conclusion

Three findings came out of the evaluation. First, the era split is necessary: GPT-4o-mini's 73.1% pre-cutoff baseline drops to 0.0% post-cutoff, so what looks like reasoning on previous meetings is mostly memorization, and retrieval's real contribution is only visible on meetings the model has not seen. Second, on date-targeted queries BM25 beats both dense retrieval and the hybrid RRF combination that the RAG literature usually treats as the safe default. Hybrid scored 75.0% post-cutoff to BM25's 91.7%, because the dense leg pulls semantically related but wrong meetings into the fused top-5. Third, accuracy alone hides retrieval failure when class distributions shift: semantic retrieval looked like it improved on recent meetings (7.7% to 58.3%) while its hit@5 stayed at 0.000 in both eras, which is why hit@5 is the trustworthy retrieval signal here.

The broader takeaway is that "hybrid is best" is not unconditional. RRF assumes the two retrievers reinforce each other. On queries where one of them has no useful representation of the query (here, dates in a dense vector space), fusion makes the answer worse. For RAG systems built on similar metadata-style queries, keyword retrieval alone, or a structured path that targets metadata fields directly, can be more reliable than a typical RRF implementation.

The biggest lesson from building this was that the evaluation design did more for the paper than any single architectural choice. The era split, the binary hit@5 metric, and the purposefully simple query template are what made the failure modes visible.

## 10. References


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**AI Use:** I used Claude (Sonnet 4.5 and Opus 4.7) as a coding and writing assistant throughout the project, including for portions of the code in src/ , and the structure of this write-up. Final design decisions, the golden-set labeling, the evaluation protocol, and the analysis of results are mine.